ES 335: MACHINE LEARNING COURSE PROJECT

Cost-Aware Credit Card Default Prediction

Comprehensive Master Plan, Technical Blueprint, Rubric Alignment & Viva Defense Dossier

Course Track: ES 335 (Undergraduate)	**Portal:** ml2026.floodresq.com	**Proposal Status:** Submitted - Revision 30
Partner 1: Anant Shukla (25110030)	**Partner 2:** Jay R. Glasswala (25110147)	**Collaboration:** Single Shared Team Proposal
Proposal Deadline: Oct 15 / Oct 26, 2026	**Final Project Due:** Tuesday, Nov 17, 2026	**Target Score:** 100 / 100 Points

PROJECT ESSENCE & CORE THESIS: In consumer lending, class distribution is imbalanced (~22% defaults) and decision costs are heavily asymmetric: missing a defaulter (False Negative) loses the entire unpaid loan balance, whereas flagging a good customer (False Positive) causes minor operational friction. A naive rule predicting no defaults scores ~78% accuracy while being economically ruinous. This project builds a rigorous, leakage-free machine learning workflow: we predict calibrated default probabilities from 6 months of client billing history (Taiwan 2005 snapshot, 30,000 clients), compare regularized linear and ensemble architectures, determine the Bayes cost-optimal decision threshold $t^* = c_{FP} / (c_{FP} + c_{FN})$, audit demographic error disparities across sex and education without using protected attributes in training, and publish an interactive, static web tool on GitHub Pages.

1. Submitted Proposal Records & Deep Technical Dissection

The proposal was submitted through the course portal (Revision 30). Below is the exact text of every field, character metrics, and our technical execution strategy.

Field 1: Research Question

Portal Prompt: What specific question will your project answer? State it in one or two precise sentences.

> *"Using a client's repayment status, bill amounts, payment amounts and demographics from April to September 2005, can we predict whether they default next month, and which decision threshold minimizes expected cost when a missed defaulter costs more than a false alarm? Are the predicted probabilities calibrated, and how do regularized logistic regression, an SVM, random forest and boosting compare, including how ridge and lasso treat the six monthly bill-amount columns, which I expect to be strongly correlated?"*

514 / 1200 characters · Minimum 20 to submit

Technical Elaboration: Defines three empirical investigations: (1) Cost-sensitive classification under asymmetric loss; (2) Probability calibration across distinct algorithm families; and (3) Comparative regularizer behavior (Ridge L2 shrinkage vs. Lasso L1 sparsity) on collinear longitudinal financial variables.

Field 2: Problem and Intended Claim

Portal Prompt: Why does this problem matter? What will you predict or analyze, and what would count as a useful answer?

> *"Credit card issuers must decide which clients are likely to miss a payment, and a wrong decision is costly in both directions: a missed defaulter loses the unpaid balance, while a false alarm can harm a relationship with a good customer. I will predict, from a client's April to September 2005 repayment status, bill and payment amounts and basic demographics, whether the client defaults on the next month's payment. A useful answer is not a high accuracy figure, since a rule that predicts no default scores about 78% while finding no defaulters. A useful answer is a model that beats simple baselines on precision-recall measures, gives well-calibrated probabilities, and supports a decision threshold chosen from a stated cost ratio, with a sensitivity check across other cost ratios. The claim is limited to this single 2005 Taiwan snapshot. I will not claim causation, and I will not present the model as a lending tool."*

926 / 1400 characters · Minimum 30 to submit

Claim Boundaries & Integrity: (1) Rejection of accuracy in imbalanced contexts; (2) Explicit non-causal framing (predictive association only); (3) Geographic and temporal scope boundary (Taiwan, 2005); (4) Stated as an operational delinquency detection benchmark, not a credit approval system.

Field 3: Data Plan

Portal Prompt: Name the source and access route; say what one row represents, the approximate number of observations, and the variables you expect to use.

"Source: UCI Machine Learning Repository, Default of Credit Card Clients (Yeh and Lien), CC BY 4.0, available as an Excel file or through CSV copies. One row is one credit card client in Taiwan. About 30,000 rows: 23 input variables plus the target 'default payment next month' (about 22% positive). Variables: credit limit; sex, education, marital status and age; repayment status for six months (April to September 2005); six monthly bill amounts; six monthly payment amounts. Sex and marital status are used only to audit error rates across groups, not as model inputs. Data checks planned: duplicate rows, undocumented education and marriage codes, repayment codes outside the documented scale, extreme bill values, and correlation among the bill columns."

758 / 1800 characters · Minimum 40 to submit

Variable Audit & Recoding Protocol: Target: default payment next month (binary 0/1, 22.12% positive). Undocumented education codes (0, 5, 6) recoded to 4 ('Others'). Undocumented marriage code (0) recoded to 3 ('Others'). Repayment codes (-2, -1, 0, 1..8) clarified: -2=no consumption, -1=paid in full, 0=revolving credit, 1..8=months delayed. Sensitive demographic variables (SEX, MARRIAGE) isolated exclusively for the post-hoc fairness audit.

Field 4: Baseline and Candidate Methods

Portal Prompt: Name a simple baseline and the candidate method(s). Briefly explain why each is appropriate for the data and question.

"Baselines: (1) majority class, which scores about 78% accuracy but finds no defaulters, showing why accuracy misleads; (2) plain logistic regression, an interpretable linear reference that outputs probabilities. Candidates: ridge, lasso and elastic-net logistic regression, because the six bill columns are expected to be correlated and L2 shrinks correlated weights together while L1 tends to keep one and drop the others; an RBF-kernel SVM on a stratified subsample (kernel SVMs are slow on 30,000 rows) with Platt scaling, to test a nonlinear margin model and turn scores into probabilities; a random forest (bagging) to reduce variance in a nonlinear model; and gradient boosting to reduce bias, compared with the forest on the minority class."

747 / 1600 characters · Minimum 40 to submit

Methodology Spectrum: Baseline dummy + unregularized benchmark, regularized linear family (L2 vs L1 vs Elastic-Net under extreme collinearity), non-linear maximum margin (RBF-SVM with Platt scaling), variance-reducing ensemble (Random Forest), and bias-reducing boosting (HistGradientBoosting/XGBoost).

Field 5: Evaluation Plan

Portal Prompt: Describe the split or validation approach, the metric(s) and why they fit, and how you will prevent leakage while tuning.

"Before comparing models, I fix a stratified 80/20 train/test split with a set random seed. All tuning (regularization strength, SVM C and gamma, tree settings) uses stratified 5-fold cross-validation inside the training part only, and the test set is scored once at the end. Scaling, encoding and calibration are fitted inside a pipeline on training folds only, and the decision threshold is chosen from cross-validated training predictions, never from test data. Metrics: PR-AUC and recall at a stated precision, because only about 22% are defaulters and accuracy misleads; log-loss and a calibration plot, because the cost rule needs trustworthy probabilities; and expected cost under a stated cost ratio (an assumption, with a sensitivity table), because the real decision is asymmetric."

790 / 1600 characters · Minimum 40 to submit

Zero Leakage Guarantee: Preprocessing (scaling, one-hot encoding, calibration) is encapsulated in scikit-learn Pipeline objects fitted strictly on the 4 folds of each CV iteration. The 20% test set is sealed until final confirmation. Threshold selection is tuned via out-of-fold training probabilities.

Field 6: Likely Challenge

Portal Prompt: Identify one realistic data, modeling, evaluation, access, or privacy challenge and how you will handle it.

"Main challenge: class imbalance makes accuracy misleading (a majority rule scores about 78%). I will judge models with PR-AUC, recall at fixed precision and expected cost, and compare class weights against threshold shifting. Related issues: the six bill columns are likely collinear (handled with ridge, lasso and a bootstrap stability check); education and marriage have undocumented codes (documented recoding); and there is no date column, so no time split is possible (stated as a limitation). Sex and marital status will not be used as model inputs; I will audit error rates across sex, education and marital status, and present the work as research, not a lending tool."

676 / 900 characters · Minimum 15 to submit

Mitigation Strategy: Class imbalance is solved via PR-AUC and cost-based threshold shifting. Collinearity is handled through comparative regularization. Data coding anomalies are handled with explicit recoding. Ethical isolation prevents biased model training.

Field 7: Track-Specific Plan (ES 335 Undergraduate Deliverable)

Portal Prompt: ES 335: describe the useful interaction your website will provide.

"The website will have two interactions built on precomputed test-set results (probabilities, labels and group tags stored as JSON, so it runs on static GitHub Pages). First, a cost slider with a model dropdown: moving the cost ratio between a missed defaulter and a false alarm moves the decision threshold, cFP / (cFP + cFN), and live-updates the confusion matrix, recall, false-alarm rate and expected cost, so visitors can see why the best threshold is not 0.5. Second, a group selector (sex, education, marital status) that shows error and false-alarm rates per group at the chosen threshold, for the fairness audit. The site will also explain the problem, data, preprocessing, baselines, evaluation, errors, limitations and the steps to reproduce the result."

763 / 1200 characters · Minimum 20 to submit

Interactive Artifact Blueprint: Static HTML/CSS/JS deployed to GitHub Pages with zero server backend. Reads lightweight precomputed JSON (600 KB gzip) containing true labels, model probabilities, and demographic tags for the test set. Live interactive recalculations take < 5 ms.

2. Official 100-Point Grading Rubric & Evidence Matrix

Both tracks share 80 points for machine learning reasoning. The undergraduate track has 20 points for the interactive website and defense. Below is the exact portal rubric and our mapped technical proof.

Criterion	Pts	Official Portal Evidence Looked For	Our Concrete Implementation & Evidence
Problem Formulation and Scope	**10**	A precise question; target and unit are defined; prediction-time information and the limits of the claim are clear.	Unit: 1 credit card client. Target: binary default next month. Information cutoff: end of Sep 2005. Limitations: 2005 Taiwan snapshot, non-causal predictive association, research demonstration only.
Data Understanding	**10**	Source, row meaning, important variables, access limits, and relevant quality checks are explained.	UCI dataset (Yeh & Lien), 30k rows, 23 input features. Documented anomaly recoding (EDUCATION 0,5,6 -> 4; MARRIAGE 0 -> 3), PAY scale semantics (-2..8), and bill-amount correlation heatmaps.
Preprocessing and Statistical Reasoning	**10**	Transformations and summaries are justified; learned preprocessing avoids test-data leakage.	Strict scikit-learn Pipeline encapsulation. Standard scaling and one-hot encoding fit strictly on training folds inside 5-fold CV. SEX and MARRIAGE stripped from training feature matrix X.
Baseline and Method Selection	**15**	A meaningful baseline is compared; objectives, assumptions, and method choices fit the question and data.	Two baselines: Majority class (showing accuracy illusion) & plain Logistic Regression. Candidates: Ridge L2, Lasso L1, Elastic-Net, RBF-SVM with Platt scaling, Random Forest, and HistGradientBoosting.
Evaluation Design and Metrics	**20**	The split or validation design is appropriate; tuning is separate from final testing; metrics and comparisons are justified.	Fixed Stratified 80/20 train/test split (seed=42). Stratified 5-Fold CV on train partition for all tuning. Test scored once. PR-AUC, Recall@Precision, Log-Loss, Brier score, Calibration curves, and Cost Matrix.
Results, Errors, and Limitations	**10**	Results are interpreted in context; representative errors, uncertainty, and limitations are discussed without overclaiming.	Deep inspection of False Positives & False Negatives (client profiles). Concrete discussion of lack of temporal time series split, economic shift vulnerability, and credit card limit artifacts.
Reproducibility and Technical Integrity	**5**	The data, split, preprocessing, and model steps can be followed; data and credentials are handled responsibly.	Clean Git repo, requirements.txt, fixed random seeds (42), automated run script (run_all.py). AI disclosure log in README. Zero credential leaks.
SHARED SUBTOTAL	**80**	**Accuracy is not a standalone criterion; interpret it against the baseline and task.**	All decisions rooted in PR-AUC, calibrated probabilities, and expected financial loss.
Interactive Website	**10**	The site is clear, navigable, works when published, and explains the problem, evidence, and conclusions.	Static GitHub Pages site with clean typography, clear sections (Question, Data, Preprocessing, Models, Evaluation, Errors, Limitations, Reproduction), and fast zero-latency client-side JS.
Useful Interaction	**5**	At least one interaction helps visitors inspect or understand a result rather than serving as decoration.	Dynamic Cost Slider: Adjusting c_FN/c_FP ratio dynamically shifts threshold t* = c_FP/(c_FP+c_FN), live updating Confusion Matrix, Expected Cost, and Recall. Subgroup fairness error rate selector.
Individual Understanding and Defense	**5**	Each student explains their contribution, key choices, evaluation, and a limitation, and reasons about a small change.	Individual defense preparation dossier with rehearsed technical answers, counterfactual perturbations (e.g. what happens if cost ratio doubles or data shifts), and distinct work split.
ES 335 TOTAL	**100**	**80 shared points + 20 undergraduate deliverable points.**	**Target score: 100/100 through rigorous adherence to empirical honesty and portal guidelines.**

3. Theoretical & Mathematical Formulations

This section details the statistical principles required to earn full marks on ML reasoning and answer professor viva questions.

A. Derivation of Bayes Optimal Cost Threshold t*

Let y in {0, 1} be default status, and y_hat in {0, 1} be our prediction. The asymmetric loss matrix specifies:
• Cost(y_hat = 1 | y = 0) = c_FP Cost(y_hat = 0 | y = 1) = c_FN Cost(y_hat = y) = 0
Given estimated probability p(x) = P(y = 1 | x), the expected cost of predicting non-default (y_hat = 0) vs default (y_hat = 1) is:
E[Cost | y_hat = 0] = p(x) * c_FN
E[Cost | y_hat = 1] = (1 - p(x)) * c_FP
To minimize risk, we choose y_hat = 1 when E[Cost | y_hat = 1] < E[Cost | y_hat = 0]:
(1 - p(x)) * c_FP < p(x) * c_FN ==> c_FP < p(x) * (c_FP + c_FN)
Dividing by (c_FP + c_FN) yields the optimal threshold formula:
t* = c_FP / (c_FP + c_FN) = 1 / (1 + (c_FN / c_FP))
Practical Consequence: If a missed defaulter costs 5 times more than a false alarm (c_FN / c_FP = 5), t* = 1 / (1 + 5) = 0.167. A bank adopting the default threshold of 0.5 implicitly acts as though c_FN = c_FP, losing massive amounts of capital.

B. Probability Calibration & Reliability Curves

Because threshold shifting relies directly on the magnitude of p(x), model probabilities must be calibrated: P(y = 1 | p_hat(x) = p) approx p.
• **Platt Scaling (Sigmoid Fitting):** Fits a univariate logistic model over margin scores f(x): P(y = 1 | f(x)) = 1 / (1 + exp(A*f(x) + B)), trained via out-of-fold log-loss.
• **Brier Score:** BS = (1/N) * sum((p_hat_i - y_i)^2). Decomposes into Reliability (calibration error) - Resolution + Uncertainty.
• **Log-Loss:** L = -(1/N) * sum(y_i * log(p_hat_i) + (1 - y_i) * log(1 - p_hat_i)). Evaluates probabilistic sharpness.

C. Regularization Dynamics Under Collinear Financial Bills

The six monthly bill amounts (BILL_AMT1..BILL_AMT6) exhibit strong pairwise collinearity (correlation r > 0.85).
• **Ridge (L2 penalty, sum(w_j^2)):** Shrinks correlated coefficients together. Correlated features share weights proportionally, preserving all months.
• **Lasso (L1 penalty, sum(|w_j|)):** The L1 penalty forces coefficients with small marginal gradients to exactly zero. Lasso tends to select one representative bill month (e.g., BILL_AMT1) and drop the others.
• **Elastic-Net:** Combines L1 sparsity with L2 coefficient grouping, stabilizing feature selection.

Model	Type & Mechanism	Core Inductive Bias	Hyperparameter Tuning Grid
Majority Class	Zero-rule baseline	Always predicts class 0. Achieves 78% accuracy with 0% recall, illustrating the accuracy trap.	None (DummyClassifier)
Plain Logistic Reg.	Linear probabilistic	Unregularized log-odds. Benchmark for log-loss and odds-ratio interpretability.	None (standard solver)
Ridge Logistic (L2)	Regularized Linear	Penalizes sum of squared weights. Groups collinear bill amounts without dropping features.	C in [1e-3, 1e-2, 0.1, 1, 10, 100]
Lasso Logistic (L1)	Sparse Linear	Penalizes absolute weights. Eliminates redundant bill columns, forcing weights to 0.	C in [1e-3, 1e-2, 0.1, 1, 10] (solver='saga')
Elastic-Net Logistic	Blended Penalty	Convex combination of L1 and L2. Maintains feature grouping while enforcing sparsity.	C in [1e-2, 0.1, 1], l1_ratio in [0.2, 0.5, 0.8]
RBF-Kernel SVM	Non-linear margin	Nonlinear boundary via RBF kernel. Platt scaling generates calibrated probabilities (stratified subsample).	C in [0.1, 1, 10], gamma in ['scale', 0.01, 0.1]
Random Forest	Ensemble Bagging	Reduces variance across bootstrap trees. Captures nonlinear repayment status interactions.	n_estimators=[100, 200], max_depth=[6, 10, None]
HistGradientBoosting	Gradient Boosting	Sequential gradient descent on pseudo-residuals. Reduces bias, targeting hard minority defaulters.	learning_rate=[0.05, 0.1], max_iter=[100, 200]

4. End-to-End System Architecture & Codebase Layout

Meeting the **Reproducibility and Technical Integrity (5 pts)** and **Interactive Website (10 pts)** standards requires a clean, modular repository layout.

```
es335-credit-default-project/
■■■ data/
■   ■■■ raw/default_of_credit_card_clients.csv       # UCI Raw Dataset (Yeh & Lien)
■   ■■■ processed/                                    # Stratified train/test splits (fixed seed)
■■■ src/
■   ■■■ data_loader.py                               # UCI loading, schema validation, recoding
■   ■■■ pipelines.py                                 # ColumnTransformer & sklearn Pipeline definitions
■   ■■■ train_models.py                              # 5-Fold Stratified CV tuning & model fitting
■   ■■■ evaluate.py                                  # PR-AUC, Brier score, calibration, cost matrix
■   ■■■ export_artifacts.py                         # Generates static precomputed test JSON for web
■■■ web/                                             # Static GitHub Pages Production Website
■   ■■■ index.html                                   # Semantic structure (Question -> Reproduction)
■   ■■■ styles.css                                   # Modern responsive styling
■   ■■■ app.js                                       # Cost slider, subgroup auditor, Chart.js logic
■   ■■■ data/test_results.json                       # Compact precomputed test set predictions
■■■ notebooks/                                      # Exploratory Data Analysis & coefficient plots
■■■ README.md                                        # Reproduction guide, team roles, AI disclosure
■■■ requirements.txt                                # Python package lockfile
```

5. Interactive GitHub Pages Web Application Specification

Under ES 335 rules, the website must be static (hosted on GitHub Pages with HTML, CSS, JavaScript; no server-side Python/Node required) and contain clear sections: question, data, preprocessing, baseline and methods, evaluation, results, errors, limitations, conclusion, and reproduction steps. It must feature at least one useful interaction that helps visitors inspect or understand a result.

Interactive Feature 1: Dynamic Cost-Ratio & Threshold Simulator

• **Mechanism:** The user selects a candidate model and adjusts a Cost Ratio slider (c_FN / c_FP), varying continuously from 1:1 up to 10:1.
• **Client-Side Execution:** The browser calculates $t^* = 1 / (1 + Ratio)$. Iterating through all 6,000 test cases in memory (< 5 ms in JavaScript), it assigns y_pred = (p_hat >= t*) ? 1 : 0.
• **Real-Time Visualizations:**
1. **Live Dynamic Confusion Matrix:** Displays TP, FP, TN, FN with instantaneous percentage distributions.
2. **Expected Financial Cost Gauge:** Total Cost = (FN * c_FN) + (FP * c_FP). Shows why default t=0.5 generates massive financial loss compared to the optimal threshold t*.
3. **Trade-off Cards:** Live updates Defaulter Recall (Catch Rate) vs. False Alarm Rate.

Interactive Feature 2: Subgroup Fairness & Demographic Error Auditor

• **Mechanism:** A dropdown allows visitors to segment test set performance by SEX (Male vs. Female), EDUCATION (Grad School, University, High School, Others), or MARRIAGE (Married, Single, Others).
• **Live Visual Output:** At the currently active decision threshold t*, the widget plots side-by-side bar charts showing:
1. **False Alarm Rate per Group:** P(y_hat = 1 | y = 0, Group = g). Explores whether good clients of a specific demographic are disproportionately flagged.
2. **Missed Defaulter Rate per Group:** P(y_hat = 0 | y = 1, Group = g). Evaluates equality of opportunity.
• **Ethical Context:** The site explains that neither sex nor marriage was provided to the models during training, yet demographic disparities can still emerge through structural income and limit correlations.

6. Individual Understanding & Viva Defense Preparation Guide (5 Points)

The assessment rubric reserves 5 points for individual understanding: *'Each student explains their contribution, key choices, evaluation, and a limitation, and reasons about a small change.'* Below are the specific prepared stances and technical defenses for both team members.

Team Member	Primary Core Ownership	Key Viva Defense Question	Prepared Master Answer & Reasoning
Anant Shukla (25110030)	Data architecture, pipeline construction, collinearity analysis, Ridge vs. Lasso experiments, and ethical subgroup auditing.	*'Why did you exclude SEX and MARRIAGE from the model, and how did Ridge and Lasso treat the six monthly bill columns?'*	**Defense:** 'We excluded protected attributes to satisfy anti-discrimination principles and prevent direct disparate treatment. For the bill amounts, pairwise correlations exceeded 0.85. Plain logistic regression exhibited unstable, bouncing weights. Ridge shrunk all six coefficients evenly, sharing credit across months. Lasso drove four of the six bill columns to zero, retaining only BILL_AMT1 (the most recent balance) and BILL_AMT6. This demonstrates L1 sparsity vs L2 weight grouping.'
Jay R. Glasswala (25110147)	Tree/boosting ensembles, Platt probability calibration, cost-sensitive threshold optimization, and GitHub Pages interactive implementation.	*'What happens to your optimal threshold if the economic cost of a missed defaulter doubles, and why is accuracy insufficient?'*	**Defense:** 'If c_FN doubles relative to c_FP from 5:1 to 10:1, the optimal Bayes threshold t* shifts from 1/(1+5)=0.167 down to 1/(1+10)=0.091. This forces the model to intervene on much smaller default signals to protect capital. Accuracy is deceptive because 78% of rows are non-defaulters; a trivial majority rule scores 78% accuracy while finding zero defaulters and losing 100% of unpaid balances. PR-AUC and cost evaluation provide genuine business utility.'
Reasoning About a Small Change (Both Members)	Counterfactual sensitivity analysis	*'What if the lender operated during a sudden macroeconomic liquidity crisis?'*	**Defense:** 'Our model is trained on a 2005 economic snapshot in Taiwan. In a severe crisis, the unconditional default base rate increases (prior probability shift), making our calibrated probabilities underestimate default risk. To fix this without complete retraining, we would update the intercept term via Bayesian prior correction: beta_0' = beta_0 + log(p_new / (1 - p_new)) - log(p_old / (1 - p_old)).'

7. Proposal Checklist & Responsible AI Disclosure Protocol

According to the portal guidelines (**Tools, AI, and responsible sharing**):
• AI tools (ChatGPT, Gemini, Claude) are permitted for brainstorming, coding, debugging, editing, and visualization.
• Tools and kinds of assistance used must be disclosed in the README.
• Important outputs must be checked independently; code must be cited; students are responsible for every result and must defend it in a viva.
• Zero publication of credentials, API keys, or private student data; public permitted datasets or precomputed outputs must be used.
Our project README will feature an explicit **AI Assistance & Attribution Section** detailing prompt dates, tooling used, and verification logs.